

Deconstructing scientific text simplification

with direct and multi-stage LLM approaches

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Task setup



- Task 1.1 sentence level, Task 1.2 document level, Cochrane-auto
- One base model throughout (GPT-4.1), one fixed pipeline
- 32 runs, single seed 123
- 2024 Cochrane-auto corpus (Bakker & Kamps, TSAR 2024)

THREE QUESTIONS

- 1 Does stage-wise decomposition beat direct one-pass rewriting?
document level
- 2 Does relevance-aligned retrieval beat random pooled sampling?
both levels
- 3 Does mined guidance beat manual prompt engineering?
both levels

Decomposition designs

Q1 - Does stage-wise decomposition beat direct one-pass rewriting?

Plan-based

stages emit plans, not prose

- Structure plan: what to split, reorder, compress
- Terminology plan: glossary of jargon
- Final stage rewrites using both

Dual-draft

each stage emits a full abstract

- Terminology draft: swap jargon only
- Structure draft: split and reorder only
- Merge resolves toward the original

Factor isolation

the drafts reused alone

- Terminology draft as final output
- Structure draft as final output
- Isolates lexical against syntactic change

Both pipelines end with a grounding check, and one repair pass on failure

Few-shot selection

Q2 - Does relevance-aligned retrieval beat random pooled sampling?

Random pooled

- Filtered candidate pool
- Quality and length filtering
- Evaluation overlap excluded

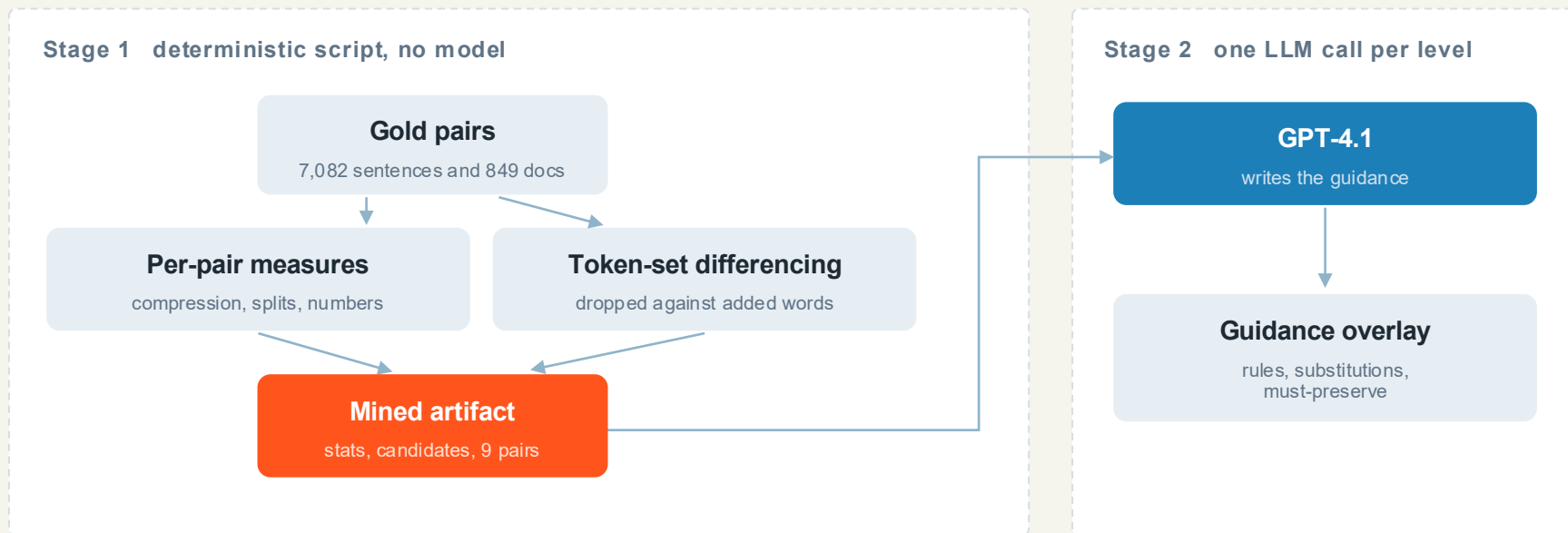
Relevance-aligned best-match

- Content overlap, Jaccard on token sets: 0.65
- Complexity profile: length, long words, digits: 0.20
- Near-duplicate filtering for diversity

Hypothesis: demonstrations help more when close to the target in content, numbers and difficulty

Guidance pipeline

Q3 - Does mined guidance beat manual prompt engineering?



The two overlays

PROMPT-GUIDED G1S

- Use short, clear sentences.
- Replace medical jargon with common words when possible.
- Keep important numbers and statistics.
- Preserve the main findings and outcomes.
- Keep uncertainty or hedging language (e.g., may, probably).
- Remove unnecessary details about study design.
- Summarize long or complex lists into simpler forms.
- Avoid technical terms for study quality unless needed.
- Use 'people' instead of 'participants' when appropriate.
- Use 'side effects' instead of 'adverse events.'
- Use 'death' or 'deaths' instead of 'mortality.'
- Do not add new information not in the original.
- Keep the meaning and intent of the original sentence.
- Use active voice when possible.
- Avoid repeating information.

STRUCTURED GUIDANCE G2S *identical rules, plus two blocks*

Must preserve:

- Main medical finding or outcome
- Key numbers and statistics
- Uncertainty or hedging language
- Any mention of side effects or harms
- The overall meaning and intent

Preferred jargon substitutions:

- participants → people included → found
- adverse → side identified → found
- adverse events → side effects
- mortality → death / deaths studies → study
- intervention → treatment outcomes → results
- Apply only when meaning remains unchanged.

Jargon survival: 0.572 non-guided → 0.300 prompt-guided → **0.233 structured**

Task 1.1 results

Sentence level · 175 plain language summaries

Run	SARI	BLEU	FKGL
Prompt-guided, FS3 best-match	42.07	12.81	8.12
Structured-guidance, FS3 best-match	41.73	12.98	8.20
Prompt-guided, zero-shot	41.47	11.76	9.71
Legacy, context + FS3 best-match	38.50	12.41	10.61
Non-guided, FS3 best-match	38.03	12.33	9.25
Legacy, sentence-only zero-shot	37.13	8.33	10.91
Non-guided, FS3 random pooled	34.94	11.18	10.06

Jargon survival falls from 0.572 to 0.233. The guidance removes the targeted complex terms rather than paraphrasing around them.

+3.09

SARI, best-match over random pooled

+4.36

SARI, guidance over non-guided

41.47

guidance with no demonstrations, above 38.03 for retrieval with no guidance

Task 1.2 results

Document level · 175 plain language summaries

Run	SARI	BLEU	FKGL	Compr.
Direct, FS3 random pooled	45.33	6.92	9.59	0.40
Structured-guidance, FS3 random pooled	45.20	4.54	9.13	0.33
Structured-guidance, FS3 best-match	44.82	4.10	8.82	0.30
Structured-guidance, zero-shot	44.31	5.50	8.87	0.40
Direct, FS3 best-match	44.25	4.38	9.69	0.31
Legacy, FS3	43.50	5.81	11.05	0.45
Legacy, zero-shot	42.98	6.29	10.42	0.50
Direct baseline, zero-shot	40.70	11.15	8.75	0.64

- Direct one-pass with diverse demonstrations wins
- Best-match hurts here, 1.08 SARI below random pooled
- The BLEU leader is the most conservative editor
- Top five sit inside 1.1 SARI

Decomposition ablations

Variant	SARI	BLEU	FKGL	Compr.	Splits	Leven.	Add.	Del.	Lex.C.
Direct, FS3 random pooled	45.33	6.92	9.59	0.40	0.61	0.43	0.10	0.71	8.42
Direct baseline, zero-shot	40.70	11.15	8.75	0.64	0.92	0.59	0.17	0.50	8.55
Decomposed, FS3 random pooled	37.80	8.99	11.66	0.89	0.95	0.63	0.30	0.39	8.49
Decomposed, zero-shot	34.67	8.38	12.12	1.27	1.43	0.68	0.41	0.22	8.49
Decomposed, FS3 best-match	33.70	8.85	12.16	1.20	1.34	0.70	0.38	0.23	8.53
Decomposed-dual, zero-shot	30.39	11.27	13.76	1.00	0.97	0.80	0.18	0.19	8.74
Terminology-only	28.42	10.09	15.75	1.17	0.98	0.80	0.27	0.14	8.69
Structure-only	22.23	11.85	11.16	0.96	1.13	0.90	0.06	0.10	8.83

Best-match against random few-shot

Task	Prompt family	SARI	BLEU	FKGL	Compression
1.1	Non-guided	+3.09	+1.15	-0.81	0.94 → 0.77
1.1	Structured-guidance	+2.43	+0.78	-0.77	0.81 → 0.69
1.2	Direct, non-guided	-1.08	-2.54	+0.10	0.40 → 0.31
1.2	Structured-guidance direct	-0.38	-0.44	-0.31	0.33 → 0.30

- Best-match compresses more in every pair. It amplifies editing.
- Document baselines already edit past the reference, so it costs SARI and BLEU.

Conclusion



- Granularity decides the winner: guided + best-match at sentence level, direct + diverse demonstrations at document level
- Retrieval amplifies editing, which helps or hurts depending on where the baseline sits
- Guidance turns near-copying baselines into reference-aligned editors
- No single run maximises SARI, BLEU and FKGL jointly



NEXT STEPS

- Embedding-based semantic matching, to catch paraphrase that token overlap misses
- Behaviour-aware retrieval on expected compression and deletion
- Sentence-level decomposition, the untested cell
- Edit-action plans instead of free-text plans



Questions?



Retrieval discriminability

Backup · Appendix A

Quantity	Sentence	Document
Best candidate, mean top-1 similarity	0.4474	0.3107
Pool average similarity	0.2464	0.2600
Gap over pool average	+81.6%	+19.5%
Top-1 range across instances	0.149 to 1.000	0.260 to 0.359
Lexical component lift	+599.5%	+59.3%
Complexity component lift	+7.6%	+1.9%

At document level the best match is barely better than a random candidate

Content overlap is the only discriminative component, and it collapses on long inputs

Evaluation settings

Backup · why our numbers differ across the organizers' tables

	Length kept	Added
Reference, 32 aligned abstracts	0.44	0.11
Reference, 175 summaries	0.79	0.34
Our best document run	0.40	0.10

Alignment filtering removes reference content with no source counterpart

Our profile matches the aligned reference, not the unaligned one

Rank improves as alignment quality increases: 15th, 10th, 8th

Caveat: 32 abstracts is a small sample, and the organizers report broadly consistent rankings across both settings

Spurious content

Backup · organizers' overgeneration analysis



Level	Our runs	Reading
Sentence	0.01	among the cleanest shown
Document	0.76 to 0.87	high on the proxy
Document, human reference	0.42	the proxy flags the gold standard too

The proxy is a token-sequence alignment: it flags output sentences left unmatched to the source

It degrades on heavily compressed output, which is why it also flags the references

Our document runs sit at the aggressive end of the field by design, compression 0.30

Counts are not comparable across teams: 175 documents for our runs against 6,928 or 8,069 for others

Semantic consistency

Backup · NLI diagnostic, source premise to output hypothesis

Level	Run	Entailment	Contradiction	Neutral
Document	Direct baseline	0.137	0.126	0.737
Document	Direct FS3 random pooled	0.154	0.097	0.749
Document	Direct FS3 best-match	0.194	0.137	0.669
Sentence	FS3 random pooled	0.788	0.015	0.197
Sentence	FS3 best-match	0.892	0.016	0.092
Sentence	Prompt-guided FS3 best-match	0.834	0.017	0.149

Sentence-pair NLI is unreliable on long, heavily compressed abstracts

Document scores do not vary monotonically with editing intensity, so they cannot rank runs

Treated as a coarse secondary check only

Mined patterns

Backup · Stage 1 output



Statistic	Sentence	Document
Training pairs	7,082	849
Median compression ratio	1.000	0.568
Numeric-token preservation	0.721	0.080
Sentence-splitting rate	0.012	0.008
Mean long-word reduction	1.02	40.95

TOP SUBSTITUTIONS

Sentence: participants → people (140), included → found (116),
adverse → side (90), mortality → death (57)

Document: confidence → found (102), adverse → side (67), reported
→ found (54), outcomes → found (51)

HEDGE RETENTION

may 0.66 probably 0.66 uncertain 0.67

unclear 0.49 low-certainty evidence 0.17

Document candidates are noisier: whole-abstract differencing pairs unrelated co-dropped and co-added tokens

Document overlays

Backup · G1D and G2D differ more than the sentence pair

Shared with the sentence overlay

Short sentences, plain words

Preserve hedging language

people instead of participants

death instead of mortality

Do not introduce new information

Document-only instructions

Reduce document length by omitting detail

Avoid raw numbers or statistics unless essential

Explain or omit statistical terms

Replace confidence interval, adverse, outcomes

Focus on who was studied and what was found

The instruction to avoid raw numbers is the opposite instinct from the sentence version, where numbers must be kept. That plus the noisy mined substitutions is why guidance helped less at document level.

Run inventory

Backup · 32 runs, single seed 123

Family	Task 1.1	Task 1.2
Legacy CLEF 2024 prompts	S1, S6	D1, D2
Enhanced direct prompting	S2, S3, S4, S5	D3, D4, D5
Prompt-guided overlay	S3+G1S, S4+G1S	D4+G1D, D5+G1D
Structured-guidance overlay	S3+G2S, S4+G2S	D4+G2D, D5+G2D
Decomposed plan-based	not run	D6, D7, D8, D9
Decomposed dual-draft	not run	D10 to D13
Factor isolation	not run	D10, D14

Few-shot retrieval varies within families: random pooled against relevance-aligned best-match, under matched prompts